# Charlotte Wong

781-728-9626 | chrlottewong@gmail.com | linkedin.com/in/eecharlottewong | github.com/sharlitwong | sharlitwong.github.io/char

## Education

**Tufts University** — Medford, MA

*Bachelor of Science in Electrical Engineering* — *Aug. 2024 – May 2028*

- GPA: 3.99/4.00 (Overall); 4.00/4.00 (EE & CS courses) • Dean's List 4/4 semesters • Year: Junior
- **Relevant Coursework:** Digital Logic Design, Embedded Systems, Electronics I (transistor circuits), Data Structures (C++), Linear Systems, Probabilistic Systems Analysis; *in progress (Fall 2026):* Computer Organization, Digital Signal Processing, Real-Time Embedded Systems

## Experience

**Ambarella** — Santa Clara, CA

*Software Architecture Intern* — *May 2026 – Aug. 2026*

- Implemented CPU and CUDA GPU back-propagation for a deformable-convolution training primitive, enabling quantization-aware training; verified with unit tests against Python-generated golden references and end-to-end MNIST training (36% to 93% test accuracy)
- Fixed a GPU race condition that lost concurrent gradient writes by implementing atomic-add and deterministic CSR-style gather paths; gather ran 2.1x faster at 512 channels (N=50 trials), matching atomic results to ∼1e-7 relative error
- Ran data-driven memory-layout tradeoff analysis (NCHW vs. NHWC, CUDA event timing): NCHW slows 7x once the working set exceeds L2 cache, but transpose overhead erases NHWC's end-to-end advantage except with very large deformation offsets

**Tufts University, Department of Electrical and Computer Engineering** — Medford, MA

*Teaching Assistant* — *Dec. 2025 – Present*

- **Digital Logic (ES4), Jan. 2026 – Present:** Debug student SystemVerilog/FPGA designs in weekly labs, covering FSMs, Karnaugh maps, blocking vs. non-blocking assignment semantics, and propagation/contamination delay
- **Embedded Systems (EE14), Dec. 2025 – May 2026:** Debugged student hardware/software designs at the register level with PlatformIO's cross debugger, oscilloscopes, and circuit-level probing; voted #1 course assistant in a student feedback survey

**Tufts Robotics Team: Pacbot** — Medford, MA

*Electrical Lead (prev. Member)* — *Aug. 2025 – Present*

- Leading bring-up of the team's new ESP32 hardware platform (migrating from STM32): wrote motor-encoder test firmware to validate the drivetrain hardware and authored an ESP32 onboarding guide for new members

## Projects

**64-bit LEGv8 Single-Cycle CPU** | *SystemVerilog, Synopsys VCS, Self-Checking Testbenches, Make* — Sept. 2026 – Present

- Designed RTL for and integrated a working 64-bit single-cycle LEGv8 (ARM) CPU executing R-type, immediate, load/store, and branch instructions (ADD/SUB, ADDI/SUBI, LDUR/STUR, CBZ, B): structural adder, ALU and ALU control, main control unit, register file, data memory, and PC
- Wrote a self-checking SystemVerilog testbench for every module: directed test cases with 4-state (`!==`) comparison so X/Z outputs are caught, `$error` diagnostics on each mismatch, and `$fatal` on failure
- Ran arithmetic, load/store, and branching test programs on the full CPU in Synopsys VCS (Makefile flows on Linux), tracing PC, registers, and memory contents in VCD waveforms to debug

**FPGA SystemVerilog Snake Game** | *SystemVerilog, iCE40 FPGA, FSM, ROM, VGA, Serial Protocol* — Oct. 2025 – Dec. 2025

- Integrated 20+ SystemVerilog modules (game FSM, PLL, VGA timing, controller interface, sprite ROMs) into a top-level design, synthesized and placed-and-routed with the open-source Yosys/nextpnr flow (apio), and brought it up on a UPDuino iCE40 FPGA
- Designed a Moore FSM driving game state with real-time self-collision and boundary checks against a shift-register snake-body model stored as a packed coordinate array
- Reverse-engineered the NES controller's bit-serial protocol, deriving latch/clock timing from a free-running counter and shifting data into a parallel register

**Wi-Fi Controlled Robot (EE31 Junior Design)** | *C++, ESP32-S3, WebSockets, Quadrature Encoders, PWM* — Sept. 2026 – Present

- Built 50 Hz closed-loop wheel-speed control using interrupt-driven quadrature encoders, feedforward duty-cycle estimates, and integral error correction, enabling fixed-radius turns through differential wheel speeds
- Wrote a non-blocking IR distance-sensor driver that cancels ambient light by subtracting LED-off from LED-on ADC readings, triggering automatic obstacle stops

## Technical Skills

**Design & Verification**: SystemVerilog, RTL Design, Self-Checking Testbenches, Synopsys VCS, Waveform Analysis (VCD), FSM Design, FPGA (iCE40, Yosys/nextpnr)
**Languages**: C, C++, Python, CUDA, ARM Assembly, Java, MATLAB
**Tools & Hardware**: Cursor (AI-assisted development), Linux, Git, Make, Oscilloscope, Multimeter, Analog Discovery 2 (WaveForms), PlatformIO, STM32, ESP32
**Spoken Languages**: Mandarin Chinese (Native), English (Native), Spanish (Working Proficiency)